v158TH FIGHTER WING
HOME OF THE GREEN MOUNTAIN BOYS
THE GREEN MOUNTAIN BOYS
HQ VERMONT AIR NATIONAL GUARD

STRIKE!

Basic Surface Attack

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Basic Surface Attack (BSA)

BSA

Basic Surface Attack, or **BSA**, is a foundational, grassroots method of AS employment against a ground target, utilizing different sets of basic flight patterns, geometry, and speeds to give a high reliability of good effects on target. Rather than reliance on rather more complex tactics or other advanced systems, BSA teaches pilot proficiency, situational awareness, and disciplined execution of core procedures.

BSA uses standardized profiles, including dive angles, release speeds, and ingress/egress altitudes to simplify decisions being made in dynamic conditions. By controlling variables like altitude, airspeed, and dive angle, pilots will achieve predictable weapons solutions against surface targets, greatly increasing the Pk and chances of accurate hits.

Beyond all, BSA is a scalable framework that can be adapted for different mission sets, including **Close Air Support (CAS)** or, more commonly in modern combat environments, **Strike Coordination and Reconnaissance (SCAR)**.

Parameters

Within any BSA employment scenario, there exists a particular set of various parameters that will ensure accurate employment of tactics and ordnance.

- **Low-Angle-High-Drag (LAHD)**
  - 2200AGL/375kts pattern
  - 10 degrees nose-low attitude
- **Low-Angle-Low-Drag (LALD)**
  - 4700AGL/350kts pattern
  - 20 degrees nose-low attitude
- **Dive Bomb (DB)**
  - 7400AGL/300kts pattern
  - 30 degrees nose-low attitude
- **High-Angle Dive Bomb (HADB)**
  - 12,300AGL/250kts pattern
  - 45 degrees nose-low attitude

Conventional Pattern

A conventional BSA pattern appears as you'd expect for a typical airfield pattern, including upwind, crosswind, downwind, and base legs. This structure offers a **predictable** and **organized flow** for aircraft stacking up to attack, making it a highly useful asset in training environments that focus on standardization and deconfliction. Conventional BSA patterns are typical in training, but rarely used in real-world tactical scenarios.

Entry into a conventional BSA pattern is typically conducted by flying directly over a target along a planned **Final Attack Heading (FAH)**. Standard entry parameters into a conventional BSA pattern consist of **350 knots**, **1,500 feet AGL** unless otherwise planned. Flying over the target, the aircraft or aircrafts performing BSA will enter a standard **overhead break** in a fingertip, route, or line-abreast formation, into the direction of the pattern.

Perched on a base-final turn, the attacker dives into the desired parameters, and should be set on the FAH no later than 2 nautical miles downwind of the target. After employment, a **Safe Escape Maneuver (SEM)** should be used.

Tactical Pattern

Unlike a conventional BSA pattern, a tactical pattern is specifically designed for flexibility, survivability, and efficiency for pilots and other assets on the battlefield. The tactical pattern does not typically adhere to any rigid flow, and is specifically designed to be fluid and adaptable to threats that could stand to be factors such as air defenses, terrain, or weather. In real-world operations where BSA is to be used, this pattern is preferred.

Entry into a tactical pattern is typically executed from a planned Initial Point, or IP, permitting the attacker to approach an area from a route that minimizes threat risks. At the IP, the attacker pops up from a low altitude configuration, and turns in to the planned FAH and dive parameters, typically a LAHD or LALD release.

After employment, a planned SEM is used, and if a re-attack is planned or requested, the attacker can remain in the pattern and continue on a REPEAT. During the entirety of the tactical pattern, timing and communication of speed, altitude, and distance are key, though especially when other aircraft are involved.

Additional Study Material

- *Basic Surface Attack* by Echo Victor
- *Conventional Pattern* by Mike Solyom
- *Tactical Pattern* by Mike Solyom
- *Basic Surface Attack* by Cider C

End

You have completed this lesson portion covering BSA.